

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users log in via email/password with securely hashed credentials (bcrypt); optional Google OAuth sign-in.	High
2	Authorization levels	Two roles: Admin (manage datasets and retrain the model) and Analyst (view / run predictions only).	Medium
3	External interfaces	Integrates with the UNDP Human Development Report API and World Bank Open Data API to retrieve indicator data.	Medium
4	Transactions processing	Each prediction request validates inputs, computes sub-indices, and returns the HDI score/tier synchronously in the same session.	High
5	Reporting	Generates downloadable PDF/CSV reports with the HDI score, tier, sub-index breakdown, and country comparison charts.	Medium
6	Business rules, etc.	HDI tier thresholds follow UNDP standard bands: Very High ≥ 0.800 , High $0.700-0.799$, Medium $0.550-0.699$, Low < 0.550 .	High
7	Compliance to laws or regulations	Complies with GDPR for any stored analyst/user data; no personally identifiable citizen-level data is processed.	Medium

8	Other	Supports model versioning so every prediction can be traced back to the specific trained model version used.	Low
---	-------	--	-----

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	A single-country prediction returns results in under 2 seconds.	Response < 2s
2	Scalability	The system supports batch CSV uploads of up to 500 countries/rows per request without degradation.	500 rows/request
3	Security & Data Privacy	All API traffic is encrypted via HTTPS/TLS 1.2+; stored credentials are hashed with bcrypt.	TLS 1.2+, bcrypt hashing
4	Reliability & Availability	The web application maintains 99% uptime during the evaluation period with automated health checks.	99% uptime
5	Usability & Accessibility	The dashboard follows WCAG 2.1 AA guidelines for color contrast and keyboard navigation.	WCAG 2.1 AA
6	Other	Codebase maintains a modular structure (data / model / API layers) for ease of maintainability.	Modular architecture